**SE 2228 Analysis & Design of Algorithms Contents (Fall 2026)**

## TOPIC OUTLINE

The **tentative** course topics are listed below. Not all topics will be covered in the same degree of detail, and the sequence may differ somewhat from the list. The main purpose of the list below is to show the material's logical structure rather than any temporal relationships. The inclusion of certain design topics depends on available time.

### A. Introduction

1. Mathematical review. (W1)
2. Problems, algorithms, instances, and programs. Types of computational tasks. (W1)

### B. Analysis of Algorithms

3. Analysis framework. (W2)
4. Empirical and theoretical analysis. (W2)
5. Asymptotic notations and basic rules. (W2)
6. Techniques and examples of algorithm analysis. (W2)
7. Analysis of various types of programming statements. (W2)
8. Analysis of non-recursive algorithms. (W3)
9. Analysis of recursive algorithms. (W3)
10. Review of sorting algorithms and their analysis. (W3)
11. More examples of algorithm analysis. (W3)

### C. Algorithm Design

12. Introduction to algorithm design. (W4)
13. Brute force. (W4)
14. Divide and Conquer. (W5)
15. Greedy algorithms. (W6)
16. Revision / Midterm. (W7)
17. Heap structures. (W8)
18. Dynamic programming. (W09 & W10)
19. Topological Sort. (W11)
20. Hashing. (W12)
21. Introduction to NP and NP-completeness. (W13)
22. Approximate algorithms for NP problems. (W13)
23. Final Review. (W14).